

Evidence of the pair-instability gap from black-hole masses

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Stellar theory predicts a forbidden range of black-hole masses between approximately $50 M_{\odot}$ and $130 M_{\odot}$ owing to pair-instability supernovae^{1–7}, but evidence for such a gap in the mass distribution from gravitational-wave astronomy has proved elusive. Early hints of a cut-off in black-hole masses at about $45 M_{\odot}$ disappeared with the subsequent discovery of more massive binary black holes^{8,9}. Here we report evidence of the pair-instability gap in LIGO–Virgo–KAGRA’s fourth Gravitational-Wave Transient Catalog (GWTC-4), with a lower boundary of $44^{+5}_{-4} M_{\odot}$ (90% credibility). Although the gap is not present in the distribution of primary masses m_1 (the bigger of the two black holes in a binary system), it appears unambiguously in the distribution of secondary masses m_2 , in which $m_2 \leq m_1$. The location of the gap lines up well with a previously identified transition in the binary black-hole spin distribution; binaries with primary components in the gap tend to spin more rapidly than those below the gap. We interpret these findings as evidence for a subpopulation of hierarchical mergers: binaries in which the primary component is the product of a previous black-hole merger and thus populates the gap. Our measurement of the location of the pair-instability gap constrains the *S*-factor for $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ at 300 keV to 260^{+190}_{-108} keV barns.

Stellar theory predicts a lack of black holes with masses from about $50 M_{\odot}$ to about $130 M_{\odot}$ (refs. 1–7). Stars with initial (zero-age main sequence) masses in the range $100\text{--}260 M_{\odot}$ are expected to experience (pulsational) pair-instability supernovae. At these masses, the carbon–oxygen stellar core is sufficiently hot that photons spontaneously produce electron–positron pairs. This leads to a drop in photon pressure that triggers sudden gravitational collapse, followed by the explosive ignition of oxygen. The resulting stellar explosion is so powerful that it can disrupt the entire stellar core in a pair-instability supernova, leaving behind no remnant whatsoever. This manifests as a gap in the black-hole mass spectrum in the $50\text{--}130 M_{\odot}$ range. Alternatively, it may trigger a series of pulses that shed enough mass such that the star collapses to a black hole below the lower edge of the gap.

Although the existence of pair-instability supernovae is a robust prediction of stellar theory, observational evidence has proved elusive. Few examples of (pulsational) pair-instability supernovae have been observed; promising candidates are the supernovae SN2018ibb and SN2020acct (refs. 10,11). Gravitational-wave observatories provide a new, promising probe of pair-instability supernovae because

they are sensitive to black holes in exactly the relevant mass range^{12–14}. Measurements of the mass gap can constrain important theoretical uncertainties, including the role of metallicity, neutrino physics, convective mixing efficiency and nuclear reaction rates⁷. For example, a driving uncertainty in the location of the gap is the uncertain $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction rate, opening up the possibility to measure this with gravitational-wave observations^{7,15}.

Previous gravitational-wave analyses have not found a clear gap in the distribution of black-hole masses^{16,17} (although see refs. 18–21, which found hints that black holes above about $40 M_{\odot}$ do not originate from stellar collapse using LIGO–Virgo–KAGRA’s third Gravitational-Wave Transient Catalog (GWTC-3)). Gravitational waves encode the primary mass m_1 and secondary mass $m_2 < m_1$ of each binary black hole. The first Gravitational-Wave Transient Catalog (GWTC-1) hinted at a cut-off in the primary mass distribution at about $45 M_{\odot}$ (for example, ref. 8), but subsequent discoveries of more massive black holes ruled out a sharp m_1 cut-off⁹. Meanwhile, an observed approximately $35 M_{\odot}$ peak in the distribution of black-hole primary mass was proposed to result from a pile-up from pulsational pair-instability supernovae^{9,22}, but subsequent

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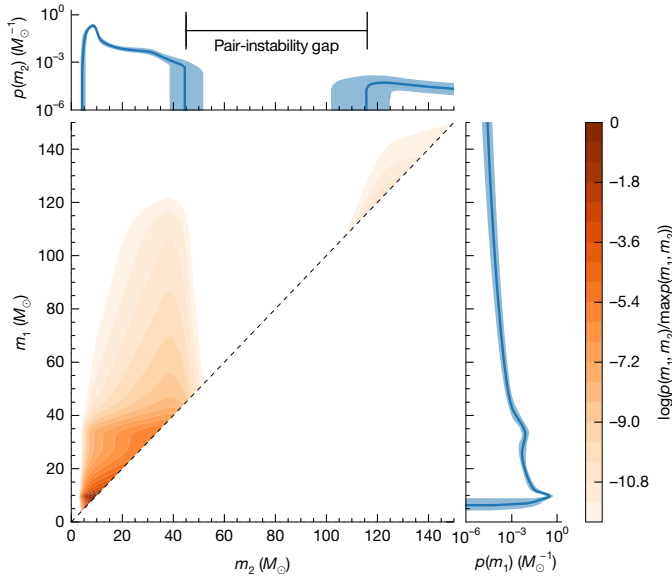


Fig. 1 | Reconstructed distribution of binary black-hole masses. The primary mass m_1 is, by definition, larger than the secondary mass, $m_2 < m_1$. The mean prediction of the joint distribution $\pi(m_1, m_2)$ is shown in orange and the median prediction of the marginal distributions $\pi(m_1)$ and $\pi(m_2)$ are shown by blue lines, with the 90% credibility range indicated by the shaded bands. The colour bar represents the two-dimensional probability density $\log(p(m_1, m_2)/\max p(m_1, m_2))$. The ‘island’ of probability at the upper right of the two-dimensional density plane is mostly associated with the high-mass event GW231123.

work suggests that this interpretation is in tension with stellar physics, observed supernovae rates and nuclear physics^{23,24}, although see, for example, ref. 25 for an alternative view.

Although the distribution of primary black-hole masses does not show a clear gap, there have not been studies looking for a gap in the secondary-mass distribution. Black holes that do not form directly from ordinary stellar collapse—such as those that originate from previous black-hole mergers²⁶, stellar mergers^{27,28}, collapsars with notable mass loss from above the gap²⁹ or which have experienced substantial accretion³⁰—may contaminate the gap. Such processes are expected to be rare and therefore unlikely to affect both black holes in a binary. We therefore analyse data from GWTC-4 (ref. 31) using a model that specifically searches for a mass gap in the distribution of m_2 . Our model allows for an interval corresponding to the pair-instability gap in m_2 , for which the probability density is zero (ref. 17); see equation (2). The absence of a gap, that is, the width of the gap to be $<20 M_\odot$, is ruled out at 99.9% credibility (Extended Data Fig. 1). The gap model is preferred over the no-gap model by a natural logarithm Bayes factor $\ln B$ of 4.4; the priors on the gap parameters as well as the details of the mass and spin models are described in Methods section ‘Basic mass-gap model’.

Figure 1 shows the black-hole mass distribution inferred under this model. Although the m_1 distribution extends unbroken from $45 M_\odot$ to $120 M_\odot$, there is a clear gap in the m_2 distribution with the lower edge at $45^{+7}_{-6} M_\odot$ (90% credibility), which is very consistent with theoretical predictions for the pair-instability gap^{1,32}. We constrain the upper edge of the gap to be $116^{+9}_{-14} M_\odot$ (90% credibility). However, the upper edge is primarily constrained by GW231123, the most massive binary detected so far³³. The upper-edge uncertainty is dominated by uncertainty in the parameters of this single event, which may or may not originate from stars massive enough to collapse to the ‘far side’ of the gap, potentially experiencing notable mass loss through disk winds and jet launching^{33,34}. By contrast, the lower edge of the gap is a robust feature of the binary black-hole population: excluding GW231123 from the analysis results in a lower edge of $43^{+7}_{-6} M_\odot$ and leaves the upper edge unconstrained. Such a gap is also present in the maximum population

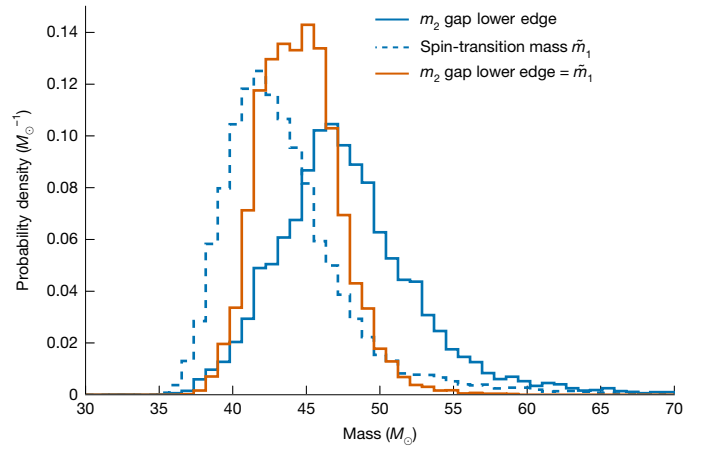


Fig. 2 | Posterior for the lower edge of the mass gap as determined by different properties. We plot the posterior on the spin-transition mass $\tilde{m}_1$ using a dashed blue line; see equation (7) for the definition. We plot the posterior on the lower edge of the gap in m_2 using a solid blue line. These two mass scales are independently but simultaneously inferred using the model allowing both m_2 gap and spin-transition mass $\tilde{m}_1$. In orange, we plot the posterior on the mass scale when the lower edge of the m_2 mass gap is taken to be the same as $\tilde{m}_1$.

likelihood distribution³⁵ (see Extended Data Fig. 4 and the discussion in Methods section ‘Comparison with the maximum population likelihood distribution’).

The presence of the pair-instability gap in the distribution of the secondary mass m_2 rather than the primary mass m_1 can be understood in the context of hierarchical mergers. Hierarchical mergers include one or more ‘second-generation’ (2G) remnants of previous black-hole mergers. Unlike ‘first-generation’ (1G) black holes that are born from stellar collapse, 2G black holes can populate the pair-instability gap. Hierarchical mergers can occur in dense stellar environments²⁶. For hierarchical mergers in dense star clusters with escape speeds typical of globular clusters, most 2G black holes are ejected from the cluster owing to the recoil kicks they receive from the anisotropic emission of gravitational waves^{36,37}. Only a small fraction are retained that can merge again with another black hole^{38,39}. The merger of two 2G black holes (2G + 2G) is expected to be comparatively rare. For example, the fraction of 2G and 1G black-hole mergers (2G + 1G) in globular clusters can be more than one order of magnitude higher than 2G + 2G mergers³⁹. Taking selection effects into account, 2G + 2G mergers could constitute at most 1% of all detected binary black-hole mergers if black holes are born with non-negligible spin³⁹, as hinted by other analyses^{16,40}. The gap that we identify in the m_2 distribution may therefore represent the dearth of 2G + 2G mergers. The prediction of 2G + 2G rates in active galactic nuclei is much more uncertain, replying on the details of gas-assisted migration.

2G black holes are expected to merge with substantial spins, inherited from the orbital angular momentum of their progenitor binaries. If they are indeed hierarchical mergers, we therefore expect the binaries with $m_1 \geq 45 M_\odot$ to include a primary with dimensionless spin of $\chi_1 \approx 0.7$. Thus, measurements of black-hole spin provide a means to independently verify our interpretation of a pair-instability mass gap contaminated with hierarchical mergers. Previous studies with GWTC-3 (refs. 20,21,41) have shown a transition of spin properties of mergers; ref. 20 first measured the transition at around the mass of $47^{+46}_{-10} M_\odot$ and ref. 42 found this transition at $m_1 = 45^{+7}_{-5} M_\odot$ and associated that with the pair-instability gap. The spin distribution of the massive binary black holes is consistent with the prediction of hierarchical mergers. To this end, we reanalyse GWTC-4 using the mass-dependent spin model developed in ref. 21. In this model, the distribution of binary black-hole effective inspiral spin χ_{eff} , a weighted sum of the black-hole spin from

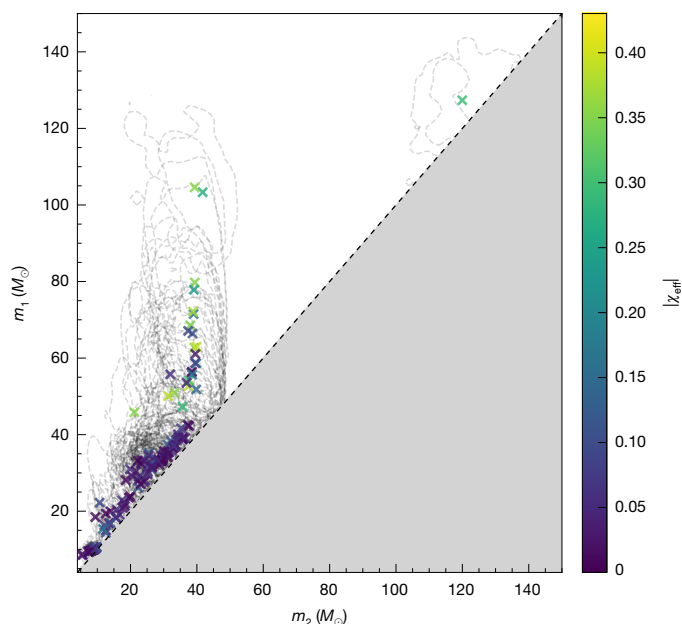


Fig. 3 | Primary masses m_1 and secondary masses m_2 coloured according to the absolute magnitude of χ_{eff} for GWTC-4 events. The masses and spins of each event are simultaneously inferred with the population. We fit the population to a model that allows for a gap in m_2 and a χ_{eff} spin distribution that transitions at the lower edge of the m_2 gap. Crosses show the median values of m_1 and m_2 . Contours show 90% credible intervals.

each component black hole, differs depending on whether the primary mass is below or above some transition mass $\tilde{m}_1$ (see Methods section ‘Spin-transition study’ for further details). Given our astrophysical interpretation of the m_2 mass gap, we expect the spin-transition mass $\tilde{m}_1$ to be consistent with the lower edge of the mass gap visible in m_2 .

The results are summarized in Fig. 2. We plot the posterior on the spin-transition mass $\tilde{m}_1$ with a dashed blue line. It is very consistent with the posterior on the lower edge of the m_2 mass gap, which is plotted with a solid blue line. We conclude that the distribution of black-hole masses and the distribution of black-hole spins provide concordant evidence for a pair-instability gap contaminated by 2G + 1G hierarchical mergers. If we force the lower edge of the m_2 mass gap to be the same as $\tilde{m}_1$, we obtain the orange posterior, which implies a lower edge of $44^{+5}_{-4} M_{\odot}$. This model, in which the spin-transition mass is fixed to the lower edge of the m_2 gap, is favoured over the transition-spin model without an m_2 gap by a $\ln B$ of 3.3 and we use it for the remaining analyses unless otherwise specified.

Using the inferred population distribution as a prior to inform the masses of individual events, Fig. 3 shows the population-informed masses m_1 and m_2 for events in GWTC-4. Each event is coloured by the posterior median of the absolute magnitude of its inspiral spin parameter $|\chi_{\text{eff}}|$. We can see a clear gap in m_2 among the observed events, together with the transition from small to large $|\chi_{\text{eff}}|$ happening around the same mass scale in m_1 as the lower edge of the gap in m_2 . Although hierarchical mergers in dense stellar clusters are usually predicted to have isotropic spin orientations owing to dynamical assembly, preferentially aligned spins can arise from black holes in active galactic nuclei disks or interactions between black holes and stars in clusters (for example, ref. 43). In Methods section ‘Symmetry of the effective spin distribution for the high-mass mergers’, we test the symmetry of the spin distribution for the high-mass mergers and show that it is consistent with being symmetric around zero, matching the prediction of 2G + 1G mergers from dense stellar clusters.

By assuming that all mergers with m_1 inside the gap are 2G + 1G hierarchical mergers, we infer a lower limit on the 2G + 1G merger rate of

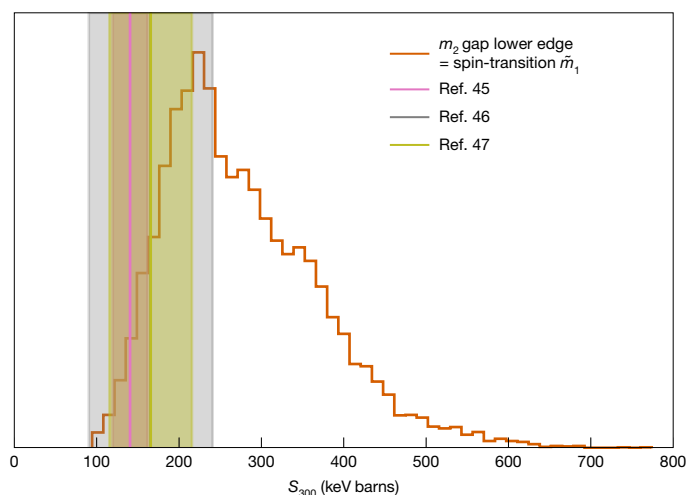


Fig. 4 | Constraints on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate. Orange is the posterior using the measurement imposing the lower edge of the m_2 gap to be equal to the spin-transition mass $\tilde{m}_1$. Pink shows the theoretical nuclear physics prediction from ref. 45 of 140^{+21}_{-21} keV barns (68% credibility). Grey shows the result from ref. 46 of 165^{+75}_{-75} keV barns (68% credibility). Yellow-green shows the result from ref. 47 of 165^{+50}_{-50} keV barns (68% credibility).

$2.5^{+2.1}_{-1.3} \times 10^{-1} \text{ Gpc}^{-3} \text{ yr}^{-1}$. We also put an upper limit on the rate of mergers with both components in the gap $< 7.9 \times 10^{-2} \text{ Gpc}^{-3} \text{ yr}^{-1}$ at 90% credibility (see Methods section ‘Gap depth investigation with a notch-filter model’ for details). By checking for events with primary masses unambiguously in the mass gap and secondary masses below the gap, we identify individual events that are likely to be 2G + 1G hierarchical mergers. Using the population-informed masses, we calculate the Bayes factor B between the hypothesis that an event contains a black hole with primary mass inside the gap to the hypothesis that it does not. We find four events with strong support for the hierarchical-merger hypothesis ($\ln B > 8$): GW190519_153544, GW190602_175927, GW191109_010717 and GW231102_071736. The exceptional event GW190521 (with total mass approximately $150 M_{\odot}$) is conspicuously absent from this list. This is because it is consistent with the ‘straddling-binary’ hypothesis⁴⁴, in which the primary mass exceeds the upper edge of the mass gap and the secondary mass falls below the lower edge, although it still has a $\ln B$ of 1.7 in favour of the primary component in the gap (see Methods section ‘Implications for GW190521’ for further details). We estimate the merger rate of straddling binaries with one component from the far side of the gap and the other component below the gap to be $5.8^{+16.6}_{-4.6} \times 10^{-3} \text{ Gpc}^{-3} \text{ yr}^{-1}$. Meanwhile, the rate of mergers with both components on the far side of the gap is $3.8^{+4.6}_{-2.2} \times 10^{-2} \text{ Gpc}^{-3} \text{ yr}^{-1}$. Assuming a GWTC-4 cumulative search sensitivity, we expect 53^{+25}_{-20} in 300 binary black-hole events to have primary masses in the gap and secondary masses below the gap. Meanwhile, we infer 6^{+7}_{-5} in 300 binary black-hole events to contain at least one component on the far side of the gap, of which 0^{+3}_{-0} can be straddling binaries. Under our best model, which indicates no detections of binaries with both components in the mass gap, Poisson uncertainty implies a 90% upper limit of $\lesssim 4.5$ such systems in a catalogue of 300 binary black-hole detections.

The location of the pair-instability gap measurement can be used to constrain the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate^{7,15,23}. In astrophysics, the reaction cross-section is commonly described by the S -factor, which is the nuclear-structure-dependent part of the cross-section for charged-particle nuclear reactions, isolated from Coulomb repulsion. Using the relationship between the lower edge of the mass gap and the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate fit from ref. 15, we constrain the astrophysical S -factor of $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ at 300 keV to 260^{+190}_{-108} keV barns (90% credibility), as seen in Fig. 4. Our constraint on the S -factor is consistent with that of ref. 45, with

the median prediction of ref. 45 within the 93% credible interval, although our posterior favours higher values, which is more consistent with the values reported, for example, in refs. 46–48. However, we do not interpret our result to be in tension with ref. 45; see Methods section ‘Constraints on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate with different models’ for more discussion.

Going forward, it is important to include the m_2 gap in binary black-hole population models. The gap can potentially be used to break the degeneracy between distance and redshift, facilitating measurements of the Hubble parameter⁴⁹. Although we interpret the dearth of binary black holes with $m_2 \geq 45 M_\odot$ as evidence of a pair-instability gap—and we consider this explanation as a promising and plausible explanation—we acknowledge that the data are open to other interpretations. For example, we could imagine that a dearth of events with $m_2 \geq 45 M_\odot$ might be produced through the complicated interplay between the initial stellar mass function and binary interactions, thereby mimicking a pair-instability gap through unrelated physical processes. If subsequent data continue to show a gap for binaries with $m_2 \geq 45 M_\odot$ regardless of the mass of the primary black hole, alternative explanations of the gap invoking binary interactions may begin to seem contrived. More excitingly, future measurements of spin orientations may help indicate the evolutionary pathways and formation environments of high-spin events to further test the hierarchical-merger hypothesis.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10359-0>.

- Fowler, W. A. & Hoyle, F. Neutrino processes and pair formation in massive stars and supernovae. *Astrophys. J. Suppl.* **9**, 201–319 (1964).
- Rakavy, G. & Shaviv, G. Instabilities in highly evolved stellar models. *Astrophys. J.* **148**, 803 (1967).
- Barkat, Z., Rakavy, G. & Sack, N. Dynamics of supernova explosion resulting from pair formation. *Phys. Rev. Lett.* **18**, 379–381 (1967).
- Fraleigh, G. S. Supernovae explosions induced by pair-production instability. *Astrophys. Space Sci.* **2**, 96–114 (1968).
- Heger, A. & Woosley, S. E. The nucleosynthetic signature of population III. *Astrophys. J.* **567**, 532–543 (2002).
- Woosley, S. E., Blinnikov, S. & Heger, A. Pulsational pair instability as an explanation for the most luminous supernovae. *Nature* **450**, 390–392 (2007).
- Farmer, R., Renzo, M., de Mink, S. E., Marchant, P. & Justham, S. Mind the gap: the location of the lower edge of the pair-instability supernova black hole mass gap. *Astrophys. J.* **887**, 53 (2019).
- Abbott, B. P. et al. Binary black hole population properties inferred from the first and second observing runs of Advanced LIGO and Advanced Virgo. *Astrophys. J. Lett.* **882**, L24 (2019).
- Abbott, R. et al. Population properties of compact objects from the second LIGO–Virgo gravitational-wave transient catalog. *Astrophys. J. Lett.* **913**, L7 (2021).
- Schulze, S. et al. 1100 days in the life of the supernova 2018ibb. The best pair-instability supernova candidate, to date. *Astron. Astrophys.* **683**, A223 (2024).
- Angus, C. R. et al. Double “acrt”: a distinct double-peaked supernova matching pulsational pair instability models. *Astrophys. J. Lett.* **977**, L41 (2024).
- Aasi, J. et al. Advanced LIGO. *Class. Quantum Gravity* **32**, 074001 (2015).
- Acernese, F. et al. Advanced Virgo: a second-generation interferometric gravitational wave detector. *Class. Quantum Gravity* **32**, 024001 (2015).
- Akutsu, T. et al. Overview of KAGRA: detector design and construction history. *Prog. Theor. Exp. Phys.* **2021**, 05A101 (2021).
- Farmer, R., Renzo, M., Mink, S., Fishbach, M. & Justham, S. Constraints from gravitational-wave detections of binary black hole mergers on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate. *Astrophys. J. Lett.* **902**, L36 (2020).
- Abbott, R. et al. Population of merging compact binaries inferred using gravitational waves through GWTC-3. *Phys. Rev. X* **13**, 011048 (2023).
- Edelman, B., Doctor, Z. & Farr, B. Poking holes: looking for gaps in LIGO/Virgo’s black hole population. *Astrophys. J. Lett.* **913**, L23 (2021).
- Mould, M., Gerosa, D. & Taylor, S. R. Deep learning and Bayesian inference of gravitational-wave populations: hierarchical black-hole mergers. *Phys. Rev. D* **106**, 103013 (2022).
- Wang, Y.-Z. et al. Potential subpopulations and assembling tendency of the merging black holes. *Astrophys. J. Lett.* **941**, L39 (2022).
- Li, Y.-J., Wang, Y.-Z., Tang, S.-P. & Fan, Y.-Z. Resolving the stellar-collapse and hierarchical-merger origins of the coalescing black holes. *Phys. Rev. Lett.* **133**, 051401 (2024).
- Antonini, F., Romero-Shaw, I. M. & Callister, T. Star cluster population of high mass black hole mergers in gravitational wave data. *Phys. Rev. Lett.* **134**, 011401 (2025).
- Talbot, C. & Thrane, E. Measuring the binary black hole mass spectrum with an astrophysically motivated parameterization. *Astrophys. J.* **856**, 173 (2018).
- Golomb, J., Isi, M. & Farr, W. M. Physical models for the astrophysical population of black holes: application to the bump in the mass distribution of gravitational-wave sources. *Astrophys. J.* **976**, 121 (2024).
- Stevenson, S. et al. The impact of pair-instability mass loss on the binary black hole mass distribution. *Astrophys. J.* **882**, 121 (2019).
- Croon, D. & Sakstein, J. Prediction of multiple features in the black hole mass function due to pulsational pair-instability supernovae. *Phys. Rev. D* **112**, 063053 (2025).
- Gerosa, D. & Fishbach, M. Hierarchical mergers of stellar-mass black holes and their gravitational-wave signatures. *Nat. Astron.* **5**, 749–760 (2021).
- Di Carlo, U. N. et al. Merging black holes in young star clusters. *Mon. Not. R. Astron. Soc.* **487**, 2947–2960 (2019).
- Renzo, M., Cantello, M., Metzger, B. D. & Jiang, Y.-F. The stellar merger scenario for black holes in the pair-instability gap. *Astrophys. J. Lett.* **904**, L13 (2020).
- Siegel, D. M. et al. “Super-kilonovae” from massive collapsars as signatures of black hole birth in the pair-instability mass gap. *Astrophys. J.* **941**, 100 (2022).
- McKernan, B., Ford, K. E. S., Lyra, W. & Perets, H. B. Intermediate mass black holes in AGN discs – I. Production and growth. *Mon. Not. R. Astron. Soc.* **425**, 460–469 (2012).
- Abac, A. G. et al. GWTC-4.0: updating the gravitational-wave transient catalog with observations from the first part of the fourth LIGO–Virgo–KAGRA observing run. Preprint at <https://arxiv.org/abs/2508.18082> (2025).
- Woosley, S. E. Pulsational pair-instability supernovae. *Astrophys. J.* **836**, 244 (2017).
- Abac, A. G. et al. GW231123: a binary black hole merger with total mass 190–265 $M_\odot$. *Astrophys. J. Lett.* **993**, L25 (2025).
- Gottlieb, O. et al. Spinning into the gap: direct-horizon collapse as the origin of GW231123 from end-to-end general-relativistic magnetohydrodynamic simulations. *Astrophys. J. Lett.* **993**, L5ss4 (2025).
- Payne, E. & Thrane, E. Model exploration in gravitational-wave astronomy with the maximum population likelihood. *Phys. Rev. Res.* **5**, 023013 (2023).
- Fitchett, M. J. The influence of gravitational wave momentum losses on the centre of mass motion of a Newtonian binary system. *Mon. Not. R. Astron. Soc.* **203**, 1049–1062 (1983).
- Gerosa, D., Hébert, F. & Stein, L. C. Black-hole kicks from numerical-relativity surrogate models. *Phys. Rev. D* **97**, 104049 (2018).
- Antonini, F. & Rasio, F. A. Merging black hole binaries in galactic nuclei: implications for advanced-LIGO detections. *Astrophys. J.* **831**, 187 (2016).
- Rodriguez, C. L. et al. Black holes: the next generation—repeated mergers in dense star clusters and their gravitational-wave properties. *Phys. Rev. D* **100**, 043027 (2019).
- Abac, A. G. et al. GWTC-4.0: population properties of merging compact binaries. Preprint at <https://arxiv.org/abs/2508.18083> (2025).
- Pierra, G., Mastroianni, S. & Perriès, S. The spin magnitude of stellar-mass black holes evolves with the mass. *Astron. Astrophys.* **692**, A80 (2024).
- Antonini, F. et al. Gravitational waves reveal the pair-instability mass gap and constrain nuclear burning in massive stars. Preprint at <https://arxiv.org/abs/2509.04637> (2025).
- McKernan, B. & Ford, K. E. S. Constraining the LVK AGN channel with black hole spins. *Mon. Not. R. Astron. Soc.* **531**, 3479–3485 (2024).
- Fishbach, M. & Holz, D. E. Minding the gap: GW190521 as a straddling binary. *Astrophys. J. Lett.* **904**, L26 (2020).
- deBoer, R. J. et al. The $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction and its implications for stellar helium burning. *Rev. Mod. Phys.* **89**, 035007 (2017).
- Buchmann, L. New stellar reaction rate for $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$. *Astrophys. J. Lett.* **468**, L127 (1996).
- Kunz, R. et al. Astrophysical reaction rate of $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$. *Astrophys. J.* **567**, 643–650 (2002).
- Shen, Y. et al. New determination of the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction rate and its impact on the black-hole mass gap. *Astrophys. J.* **945**, 41 (2023).
- Farr, W. M., Fishbach, M., Ye, J. & Holz, D. E. A future percent-level measurement of the Hubble expansion at redshift 0.8 with advanced LIGO. *Astrophys. J. Lett.* **883**, L42 (2019).

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Methods

Hierarchical Bayesian inference details

We perform hierarchical Bayesian inference with GWPopulation⁵⁰ on the recently released cumulative GWTC-4 (refs. 31,51–55). There are roughly 80 confident binary black holes detected in the first part of the LIGO–Virgo–KAGRA’s fourth observing run (O4a), enabled by a variety of detector improvements^{56–60}. Our dataset includes 153 binary black holes with false-alarm rates $\leq 1 \text{ yr}^{-1}$, consistent with ref. 40. Selection effects are taken into account by a Monte Carlo estimation^{61–64}. We use the posterior samples consistent with the choice of ref. 40, generated using Bilby and Dynesty^{65–67}. For events detected before O4a, we use the MIXED samples reported in GWTC-3 and GWTC-2.1 (refs. 68,69).

As the previously known event with the highest support for pair-instability gap components GW190521, ref. 70 concluded that the NRSur7dq4 waveform model is most faithful to numerical relativity simulations in the parameter range relevant for this exceptional event. We tested different waveform models for this event and concluded that the results of our pair-instability gap measurement are not sensitive to the choice of waveform. Using NRSur7dq4 posterior samples of GW190521, we found the lower edge of $49_{-6}^{+10} M_{\odot}$ from the mass distribution and $45_{-3}^{+5} M_{\odot}$ by incorporating spin-transition information.

Basic mass-gap model

To model the predicted pair-instability gap^{1–7,32,71–75}, we use an extension of the basic binary black-hole mass models in ref. 40. The mass model is parameterized by means of source-frame primary mass m_1 and secondary mass $m_2 < m_1$, so we are fitting the distribution

$$\pi(m_1, m_2 | \Lambda) = \pi(m_1 | \Lambda) \pi(m_2 | m_1, \Lambda), \quad (1)$$

which is conditioned on hyperparameters Λ that control the shape of the distribution. As discussed below, we use various models for $\pi(m_1 | \Lambda)$ from ref. 40. The key ingredient, however, is our model for secondary mass, which enforces a gap in the distribution of m_2 :

$$\pi(m_2 | m_1, \Lambda) \propto \begin{cases} 0 & m_g \leq m_2 \leq m_g + w_g \\ m_2^{\beta_q} & \text{otherwise.} \end{cases} \quad (2)$$

Here m_g is the lower boundary of the m_2 mass gap and w_g is the width of the gap. We assume a Bayesian prior on m_g that is uniform on the interval $(20 M_{\odot}, 150 M_{\odot})$, whereas w_g is uniform on the interval $(0 M_{\odot}, 150 M_{\odot})$. This model implies that the secondary mass is a power-law distribution unless m_2 is on the interval $(m_g, m_g + w_g)$, in which no mergers are allowed.

We perform the analysis using the broken power law + two peaks model for $\pi(m_1)$ from ref. 40. To investigate the dependence of our results on this model, we then repeat the analysis using the single power law + two peaks model from ref. 40. For our initial analysis—before we model the spin-transition mass from ref. 21 (below)—we use the default spin model from ref. 40, which consists of independent and identical truncated Gaussian distributions for the spin magnitudes of the primary and secondary black holes. Meanwhile, the cosine of the spin-tilt distribution is modelled as a mixture of a uniform distribution and a Gaussian distribution with free mean and width. The evolution of the merger rate over redshift is modelled as a power-law distribution^{9,51}.

Although the broken power law + two peaks model is found to be statistically favoured in ref. 40, the break mass is not constrained at all when we introduce the flexibility of a gap in the m_2 distribution. The slopes of the two bands of the broken-power-law distribution are consistent with each other. We interpret the initial requirement of a second power law in the m_1 distribution in present data as a projection

of the dearth of high- m_2 events. Instead, we find that the single power law + two peaks model has the highest Bayesian evidence when we allow for a gap in the secondary mass distribution. When we use the broken power law + two peaks model, the existence of the m_2 mass gap in the model is preferred over the no-gap model by a $\ln \mathcal{B}$ of 3.7. Using the single power law + two peaks model, the model with a gap is preferred with a $\ln \mathcal{B}$ of 4.4. The plots in this paper use our best mass model, single power law + two peaks with a gap, unless otherwise stated. We summarize the model comparison result in Extended Data Table 1.

Extended Data Fig. 1 shows the posterior distribution for the lower edge of the gap m_g and the upper edge of the gap $m_g + w_g$. The results from GWTC-4, including the exceptionally massive event GW231123 (ref. 33), are shown in blue. The inclusion of GW231123 is necessary to obtain a useful constraint on the upper edge of the gap. However, as noted previously, the large spins of GW231123 may suggest that it is a contamination event from a 2G + 2G merger as opposed to a binary with black holes formed above the pair-instability gap. The constraints on the m_g obtained without GW231123 (orange) are not very different from the results obtained with GW231123. The results obtained from GWTC-3 (ref. 69) are consistent with the results from GWTC-4, albeit with decreased statistical significance.

We perform a posterior predictive check^{76,77}. The distribution predicted by our mass-gap model is consistent with the observed gravitational-wave events, as shown in Extended Data Fig. 2. The cumulative density of secondary masses m_2 reaches a plateau around the lower edge of the gap at about $45 M_{\odot}$ until roughly $116 M_{\odot}$, with a tiny bump for both the observed and the predicted distributions owing to the mergers at the far end of the gap.

Gap depth investigation with a notch-filter model

We replace the model of a fixed empty gap in equation (2) by a notch filter $n(m_2)$ to investigate the depth of the gap:

$$\pi(q | \Lambda) \propto q^{\beta_q} n(m_2). \quad (3)$$

The expression for the notch filter can be written as⁷⁸

$$n(m_2) = 1 - \frac{A}{\left(1 + \left(\frac{m_g}{m_2}\right)^{\eta_{\text{low}}}\right) \left(1 + \left(\frac{m_2}{m_g + w_g}\right)^{\eta_{\text{high}}}\right)}, \quad (4)$$

in which m_g and w_g are hyperparameters describing the lower edge and the width of the gap, η_{low} and η_{high} set the sharpness of the gap’s edges and the amplitude of the gap is determined by the parameter A . We choose the same priors for m_g and w_g as our standard gap model. We use uniform priors on A in the range $[0, 1]$ as well as η_{low} and η_{high} in the range $[0, 50]$. Higher values of η_{low} and η_{high} correspond to sharp low and high edges, respectively.

We show the posteriors of those parameters in Extended Data Fig. 3. The posterior of A is sharply peaked at 1, which represents the preference of an empty gap with the lower edge of the gap consistent with our default model result. The inference of the upper edge in this notch-filter model is much more uncertain given a more flexible description of the gap, which is expected, as it is mostly driven by the detection of a single event. Non-sharp edges are disfavoured, in particular for the lower end, and increasing sharpnesses are not distinguishable (also from a hard cut-off) given the present dataset.

With the flexibility of a finite depth of the gap in this model, we can also place an upper limit on the rate of binary black holes that have both components inside the gap by the non-detection of such events. Assuming that any component inside the gap is a 2G black hole, we place constraints on the upper limit of the merger rates of 2G + 2G black holes in the gap to be $< 7.9 \times 10^{-2} \text{ Gpc}^{-3} \text{ yr}^{-1}$ at 90% credibility.

Comparison with the maximum population likelihood distribution

The π formalism introduced in ref. 35 offers a data-driven approach to visualizing the population properties of merging binaries. Rather than fitting a model, the π formalism identifies the unique distribution that maximizes the population likelihood over all possible population models. The resulting distribution $\pi(\theta)$ is always given by a weighted sum of delta functions, with the locations and weights of the ‘ π samples’ determined through constrained numerical optimization. The samples show which population features are supported by the data, as opposed to features that might be inferred as a result of model dependence. In the limit at which the number of observations tends to infinity, the π samples reproduce the true distribution³⁵.

Extended Data Fig. 4 shows π in (m_1, m_2) for the data in GWTC-4. The background contour shows the probability density inferred by the standard parametric model, single power law + two peaks with m_2 gap. The π samples are consistent with our fit; there is a notable absence of π samples with secondary mass between approximately $45 M_\odot$ and $116 M_\odot$. The π gap is not imposed by any modelling assumptions but rather emerges organically from the data. The agreement between the model-free π distribution and the parametric model analysis gives us further confidence in the existence of a pair-instability mass gap in the distribution of secondary component masses.

Mass model misspecification test

As a further test of model misspecification, we also run the analysis using the pairing-mass model from ref. 79. In this model, the joint distribution for m_1 and m_2 is written as

$$\pi(m_1, m_2) \propto \pi(m_1|A_1)\pi(m_2|A_2)f(q|\beta_q). \quad (5)$$

The hyperparameters A_1 control the shape of a nominal distribution for primary mass, whereas the hyperparameters A_2 control the shape of a nominal distribution of secondary mass (we use the word ‘nominal’ here because the marginal distribution $\int dm_2 \pi(m_1, m_2)$ is not equal to $\pi(m_1|A_1)$). The pairing function $f(q|\beta_q)$ is a power law in mass ratio $q = m_2/m_1$. With more parameters than our single power law + two peaks model, this pairing-mass model provides much more flexibility to reconstruct distributions in the m_1 – m_2 plane.

In our version of the pairing-mass model, we use separate broken power law + two peaks models for $\pi(m_1|A_1)$ and $\pi(m_2|A_2)$. For the sake of simplicity of a test focusing on the lower edge of the gap, we exclude GW231123. We also allow for the presence of different maximum truncation for m_1 and m_2 . The reconstructed distributions are shown in Extended Data Fig. 5. Notably, the pairing-mass model does not require hard cut-off in the m_2 distribution around $45 M_\odot$. This is because it can produce a dearth of events with $m_2 \gtrsim 45 M_\odot$ with a steeply falling power law. This is evidenced by comparing upper quantiles for the pairing mass and single power law + two peaks mass models. In Extended Data Fig. 6, we plot the 99th percentile $m_2^{99\%}$ of secondary masses in the population. The two distributions from the single power law + two peaks model and the pairing-mass model agree. Repeating this exercise for the 99.9% quantile (dashed lines), we see that the two distributions again agree. We conclude that the pairing-mass model fit is functionally equivalent to one with a hard cut-off in m_2 , even though this can be achieved without an explicit cut-off. The difference in the $m_1 > 175 M_\odot$ region between the models is caused by a fixed maximum $m_1 = 300 M_\odot$ truncation in our single power law + two peaks model following the set-up of ref. 40. However, the maximum m_1 truncation is not constrained at all when it is analysed as a free parameter. The choice of a fixed maximum m_1 truncation makes no difference in the inference of other notable structures of the mass distribution.

Spin-transition study

Next we replace the default spin model with the model from ref. 21, which showed evidence for a change in the spin distribution for primary masses above $\tilde{m}_1$. In our study allowing an m_2 gap, we consider two model variations of the effective inspiral spin χ_{eff} ,

$$\chi_{\text{eff}} \equiv \frac{\cos(\text{tilt}_1)\chi_1 + q\cos(\text{tilt}_2)\chi_2}{1+q}. \quad (6)$$

In the first variation, the effective inspiral spin parameter χ_{eff} is normally distributed when $m_1 < \tilde{m}_1$ and uniformly distributed when $m_1 > \tilde{m}_1$ (ref. 21):

$$\pi(\chi_{\text{eff}}|m_1, A) = \begin{cases} \mathcal{N}(\chi_{\text{eff}}|\mu, \sigma) & (m_1 < \tilde{m}_1) \\ \mathcal{U}(\chi_{\text{eff}}|w = 0.47) & (m_1 \geq \tilde{m}_1). \end{cases} \quad (7)$$

The second variation uses a mixture model for binaries with $m_1 > \tilde{m}_1$ (ref. 21):

$$\pi(\chi_{\text{eff}}|m_1, A) = \begin{cases} \mathcal{N}(\chi_{\text{eff}}|\mu, \sigma) & (m_1 < \tilde{m}_1) \\ \zeta \mathcal{U}(\chi_{\text{eff}}|w) + (1-\zeta)\mathcal{N}_u(\chi_{\text{eff}}|\mu_u, \sigma_u) & (m_1 \geq \tilde{m}_1). \end{cases} \quad (8)$$

We show in Extended Data Fig. 7 that both models produce similar results. However, the complexity of equation (8) means that this model is disfavoured owing to an Occam penalty. We therefore report our results using the spin model equation (7) throughout the paper, unless otherwise specified. As ref. 21 showed, the transition-spin model (without an m_2 gap) is strongly favoured over a model in which the entire population is represented by a single Gaussian in χ_{eff} by $\ln B > 9$. Our fiducial model, in which the spin-transition mass is equal to the lower edge of the m_2 gap, is further favoured over the model in ref. 21 ($\ln B \approx 3.3$), as well as a model in which the spin-transition mass can be different from the lower edge of the m_2 gap ($\ln B \approx 1.0$).

Reference 80 further investigates this spin-transition feature using a non-parametric model for the spin distribution of the high-mass subpopulation. It is shown that, with a weaker assumption on the spin model, the measurement of the spin transition mass $\tilde{m}_1$ is still consistent with the results using the parametric models introduced above.

Symmetry of the effective spin distribution for the high-mass mergers

We explore whether the spin distribution for the high-mass mergers is isotropic or not by a more flexible model following refs. 42,80. We extend the model in equation (7) to allow the minimum and maximum truncation of the uniform distribution that describes the spin distribution for the high-mass mergers to be free parameters.

$$\pi(\chi_{\text{eff}}|m_1, A) = \begin{cases} \mathcal{N}(\chi_{\text{eff}}|\mu, \sigma) & (m_1 < \tilde{m}_1) \\ \mathcal{U}(\chi_{\text{eff}}|\chi_{\text{min}}^{\text{high-mass}}, \chi_{\text{max}}^{\text{high-mass}}) & (m_1 \geq \tilde{m}_1). \end{cases} \quad (9)$$

We set the prior for $\chi_{\text{min}}^{\text{high-mass}}$ that is uniform on the interval $(-1, 1)$ and $\chi_{\text{max}}^{\text{high-mass}}$ that is uniform on the interval $(-1, 1)$.

We show the result of the spin distribution for high-mass mergers in Extended Data Fig. 8. Although the lower boundary of the spin distribution is relatively less constrained than the upper boundary, the overall distribution is consistent with a spin distribution symmetric around zero. We found the mean $\langle \chi_{\text{eff}} \rangle = (\chi_{\text{max}}^{\text{high-mass}} + \chi_{\text{min}}^{\text{high-mass}})/2 = -0.00^{+0.12}_{-0.10}$ and the half-width $w_{\chi_{\text{high-mass}}} = (\chi_{\text{max}}^{\text{high-mass}} - \chi_{\text{min}}^{\text{high-mass}})/2 = 0.52^{+0.12}_{-0.15}$. A positive minimum truncation, $\chi_{\text{min}}^{\text{high-mass}} > 0$, is ruled out at 99.3% credibility.

Constraints on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate with different models

On the basis of simulations, ref. 15 fits the lower edge of the mass gap as a function of the temperature-dependent uncertainty in the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate and reported constraints on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate using the first ten gravitational-wave detections. However, to determine the lower edge of the mass gap with so few events, the analysis had to use a model with strong assumptions about the shape of the black-hole-mass distribution. Following the observation of further binary black-hole signals, it became apparent that this early model was misspecified and so the earliest inferences on the lower edge of the mass gap were unreliable^{9,16}. A more recent analysis²³ explored the possibility of the bump around $35 M_{\odot}$ in the binary black-hole-mass distribution to be a signature from the pulsational pair-instability process and found the inferred astrophysical S -factor in conflict with ref. 45, illustrating the difficulty of placing constraints on $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction rates using previous gravitational-wave catalogues.

Here we show the constraints on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate with other model assumptions as well as the one in the main text. As mentioned, we used the relationship between the lower edge of the mass gap and the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate fit from ref. 15. As shown in Extended Data Fig. 9, we constrain the astrophysical S -factor of $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ at 300 keV to 243_{-119}^{+331} keV barns (90% credibility) assuming the default spin model and to 260_{-108}^{+190} keV barns (90% credibility) by incorporating spin-transition information. In blue, we also show the constraint to be 186_{-88}^{+282} keV barns using the mass-gap measurement that assumes that the spin-transition mass is independent of the m_2 gap. The measurements of the mass gap shift slightly given different assumptions on spin models owing to the degeneracy, which leads to different but consistent posteriors of the astrophysical S -factor.

Because the mapping from the lower edge of the gap to the corresponding S -factor is nonlinear, small shifts towards a lower inferred edge lead to a long tail towards higher inferred S -factors. We see that using a different model that shifts the lower edge by about $2 M_{\odot}$ results in a S -factor estimate that aligns well with the result of ref. 45 (see the blue curve of Extended Data Fig. 9). Moreover, the simulations mapping from the gap to the corresponding $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate in ref. 15 simulated the lowest edge at approximately $40 M_{\odot}$, whereas our constraint has an error bar extending to $<40 M_{\odot}$. Thus, the long tail to high S -factors in our posterior may be unrealistic.

Implications for GW190521

The presence of an apparent gap owing to pair-instability supernovae has interesting implications for previously identified gravitational-wave events with exceptional properties. GW190521 was highlighted as a potential candidate of hierarchical mergers in the mass gap^{70,81}. Although this interpretation is consistent with our results, we also find support for the hypothesis—first put forward in ref. 44—that GW190521 is a straddling binary with component black holes on either side of the gap. We show a corner plot for the mass and spin properties of GW190521 in Extended Data Fig. 10. In the context of our population model^{82,83}, the secondary black hole in GW190521 is most likely below the pair-instability gap: $m_2 = 40.7_{-7.0}^{+8.9} M_{\odot}$ at 90% credibility, although there is still marginal support for m_2 at the other end of the gap at approximately $120 M_{\odot}$. There is posterior support of about 4% for the hypothesis that both components of GW190521 are on the far side of the gap. Excluding GW231123, the overall population has a maximum $m_2 \approx 43 M_{\odot}$, as shown in orange in Extended Data Fig. 1. Thus the event GW190521 in this case is fully consistent with its secondary component below the gap.

For future work, we foresee a global fit using a model that includes subpopulations of $1\text{G} + 1\text{G}$, $2\text{G} + 1\text{G}$ and $2\text{G} + 2\text{G}$ mergers; see, for example, refs. 18,19,84–86. By explicitly modelling the $2\text{G} + 1\text{G}$ subpopulation, it would be possible to account for contamination from hierarchical mergers in the m_1 mass gap. As the number of detections increases, it will be possible to gain new insights into the pair-instability gap and the prevalence of hierarchical mergers in merging binaries.

Data availability

The results of the analyses in this work can be found on Zenodo at <https://doi.org/10.5281/zenodo.18222409> (ref. 87). The posterior samples of GWTC-4 events used in the hierarchical Bayesian inference of this work are available on Zenodo as part of LIGO–Virgo–KAGRA’s GWTC-4 data release (<https://doi.org/10.5281/zenodo.16053484>)⁸⁸. For events in GWTC-2.1 and GWTC-3, we used samples from Zenodo found at <https://doi.org/10.5281/zenodo.6513631> (ref. 89) and <https://doi.org/10.5281/zenodo.5546663> (ref. 90). We used the cumulative search sensitivity file on Zenodo found at <https://doi.org/10.5281/zenodo.16740128> (ref. 91).

Code availability

The code for this study is publicly available at https://github.com/Hui-Tong5/PISN_mass_gap_GWTC-4 as an implementation of the population models developed in this study using GWPopulation⁵⁰.

50. Talbot, C., Farah, A., Galadage, S., Golomb, J. & Tong, H. GWPopulation: hardware agnostic population inference for compact binaries and beyond. *J. Open Source Softw.* **10**, 7753 (2025).
51. Fishbach, M., Holz, D. E. & Farr, W. M. Does the black hole merger rate evolve with redshift? *Astrophys. J. Lett.* **863**, L41 (2018).
52. Thrane, E. & Talbot, C. An introduction to Bayesian inference in gravitational-wave astronomy: parameter estimation, model selection, and hierarchical models. *Publ. Astron. Soc. Aust.* **36**, e010 (2019).
53. Mandel, I., Farr, W. M. & Gair, J. R. Extracting distribution parameters from multiple uncertain observations with selection biases. *Mon. Not. R. Astron. Soc.* **486**, 1086–1093 (2019).
54. Vitale, S., Gerosa, D., Farr, W. M. & Taylor, S. R. in *Handbook of Gravitational Wave Astronomy* (eds Bambi, C. et al.) 1–160 (Springer, 2022); https://doi.org/10.1007/978-981-15-4702-7_45-1
55. Abac, A. G. et al. GWTC-4.0: methods for identifying and characterizing gravitational-wave transients. Preprint at <https://arxiv.org/abs/2508.18081> (2026).
56. Buikema, A. et al. Sensitivity and performance of the Advanced LIGO detectors in the third observing run. *Phys. Rev. D* **102**, 062003 (2020).
57. Ganapathy, D. et al. Broadband quantum enhancement of the LIGO detectors with frequency-dependent squeezing. *Phys. Rev. X* **13**, 041021 (2023).
58. Jia, W. et al. Squeezing the quantum noise of a gravitational-wave detector below the standard quantum limit. *Science* **385**, 1318–1321 (2024).
59. Capote, E. et al. Advanced LIGO detector performance in the fourth observing run. *Phys. Rev. D* **111**, 062002 (2025).
60. Soni, S. et al. LIGO detector characterization in the first half of the fourth observing run. *Class. Quantum Gravity* **42**, 085016 (2025).
61. Tiwari, V. Estimation of the sensitive volume for gravitational-wave source populations using weighted Monte Carlo integration. *Class. Quantum Gravity* **35**, 145009 (2018).
62. Farr, W. M. Accuracy requirements for empirically measured selection functions. *Res. Notes AAS* **3**, 66 (2019).
63. Essick, R. & Farr, W. Precision requirements for Monte Carlo sums within hierarchical Bayesian inference. Preprint at <https://arxiv.org/abs/2204.00461> (2022).
64. Essick, R. et al. Compact binary coalescence sensitivity estimates with injection campaigns during the LIGO–Virgo–KAGRA Collaborations’ fourth observing run. *Phys. Rev. D* **112**, 102001 (2025).
65. Ashton, G. et al. Bilby: a user-friendly Bayesian inference library for gravitational-wave astronomy. *Astrophys. J. Suppl. Ser.* **241**, 27 (2019).
66. Romero-Shaw, I. M. et al. Bayesian inference for compact binary coalescences with BILBY: validation and application to the first LIGO–Virgo gravitational-wave transient catalogue. *Mon. Not. R. Astron. Soc.* **499**, 3295–3319 (2020).
67. Speagle, J. S. DYNESTY: a dynamic nested sampling package for estimating Bayesian posteriors and evidences. *Mon. Not. R. Astron. Soc.* **493**, 3132–3158 (2020).
68. Abbott, R. et al. GWTC-2.1: deep extended catalog of compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run. *Phys. Rev. D* **109**, 022001 (2024).
69. Abbott, R. et al. GWTC-3: compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run. *Phys. Rev. X* **13**, 041039 (2023).
70. Abbott, R. et al. Properties and astrophysical implications of the $150 M_{\odot}$ binary black hole merger GW190521. *Astrophys. J. Lett.* **900**, L13 (2020).
71. Woosley, S. E. The evolution of massive helium stars, including mass loss. *Astrophys. J.* **878**, 49 (2019).
72. Marchant, P. et al. Pulsational pair-instability supernovae in very close binaries. *Astrophys. J.* **882**, 36 (2019).
73. Woosley, S. E. & Heger, A. The pair-instability mass gap for black holes. *Astrophys. J. Lett.* **912**, L31 (2021).
74. Renzo, M. & Smith, N. Pair-instability evolution and explosions in massive stars. Preprint at <https://arxiv.org/abs/2407.16113> (2024).
75. Spera, M. & Mapelli, M. Very massive stars, pair-instability supernovae and intermediate-mass black holes with the SEVN code. *Mon. Not. R. Astron. Soc.* **470**, 4739–4749 (2017).
76. Romero-Shaw, I. M., Thrane, E. & Lasky, P. D. When models fail: an introduction to posterior predictive checks and model misspecification in gravitational-wave astronomy. *Publ. Astron. Soc. Aust.* **39**, e025 (2022).
77. Miller, S. J., Ko, Z., Callister, T. & Chatzioannou, K. Gravitational waves carry information beyond effective spin parameters but it is hard to extract. *Phys. Rev. D* **109**, 104036 (2024).

78. Fishbach, M., Essick, R. & Holz, D. E. Does matter matter? Using the mass distribution to distinguish neutron stars and black holes. *Astrophys. J. Lett.* **899**, L8 (2020).
79. Farah, A. M., Fishbach, M. & Holz, D. E. Two of a kind: comparing big and small black holes in binaries with gravitational waves. *Astrophys. J.* **962**, 69 (2024).
80. Antonini, F., Callister, T., Dosopoulou, F., Romero-Shaw, I. & Chattopadhyay, D. Inferring the pair-instability mass gap from gravitational wave data. *Phys. Rev. D* **112**, 063040 (2025).
81. Abbott, R. et al. GW190521: a binary black hole merger with a total mass of $150 M_{\odot}$. *Phys. Rev. Lett.* **125**, 101102 (2020).
82. Galadage, S., Talbot, C. & Thrane, E. Gravitational-wave inference in the catalog era: evolving priors and marginal events. *Phys. Rev. D* **102**, 083026 (2020).
83. Fishbach, M., Farr, W. M. & Holz, D. E. The most massive binary black hole detections and the identification of population outliers. *Astrophys. J. Lett.* **891**, L31 (2020).
84. Doctor, Z., Wysocki, D., O'Shaughnessy, R., Holz, D. E. & Farr, B. Black hole coagulation: modeling hierarchical mergers in black hole populations. *Astrophys. J.* **893**, 35 (2020).
85. Kimball, C. et al. Black hole genealogy: identifying hierarchical mergers with gravitational waves. *Astrophys. J.* **900**, 177 (2020).
86. Kimball, C. et al. Evidence for hierarchical black hole mergers in the second LIGO–Virgo gravitational wave catalog. *Astrophys. J. Lett.* **915**, L35 (2021).
87. Tong, H. et al. Evidence of the pair instability gap in the distribution of black hole masses data release. *Zenodo* <https://doi.org/10.5281/zenodo.18222409> (2026).
88. LIGO Scientific Collaboration, Virgo Collaboration & KAGRA Collaboration. GWTC-4.0: parameter estimation data release. *Zenodo* <https://doi.org/10.5281/zenodo.16053484> (2025).
89. LIGO Scientific Collaboration & Virgo Collaboration. GWTC-2.1: deep extended catalog of compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run - parameter estimation data release. *Zenodo* <https://doi.org/10.5281/zenodo.6513631> (2022).
90. LIGO Scientific Collaboration, Virgo Collaboration & KAGRA Collaboration. GWTC-3: compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run — parameter estimation data release. *Zenodo* <https://doi.org/10.5281/zenodo.5546663> (2021).
91. LIGO Scientific Collaboration, Virgo Collaboration & KAGRA Collaboration. GWTC-4.0 cumulative search sensitivity estimates. *Zenodo* <https://doi.org/10.5281/zenodo.16740128> (2025).

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Author contributions The analyses in this work were carried out by H.T. M.F. and E.T. served as H.T.'s supervisors, providing guidance and assisting with the writing of the manuscript. Further support was provided by M.M., T.A.C., A.M.F. and S.B., who provided suggestions for analyses and helped interpret the results. N.G. performed the maximum population likelihood distribution analysis. M.M. contributed to the initial code development. S.G. helped with plotting. D.B.-M., B.F., S.G., J.G., J.H., M.K., S.J.M. and A.V. contributed to preliminary population inference that motivated this study. All authors contributed to editing the manuscript.

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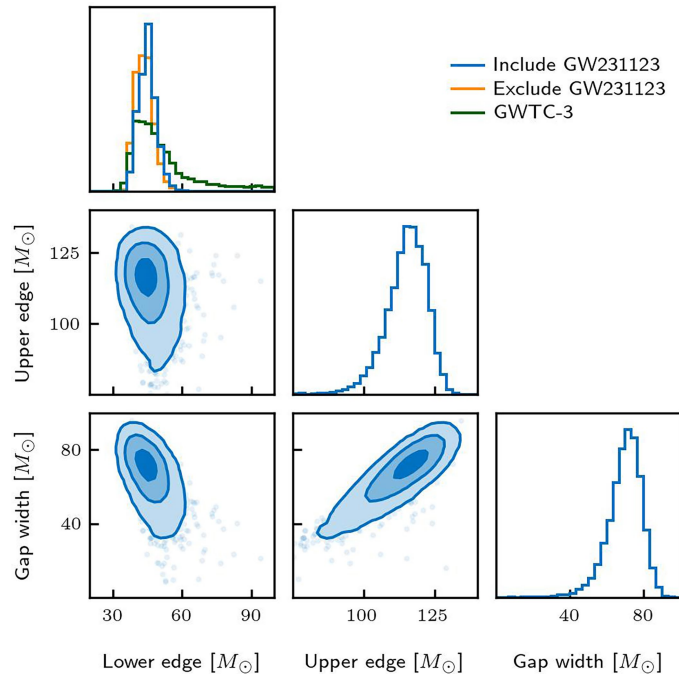
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10359-0>.

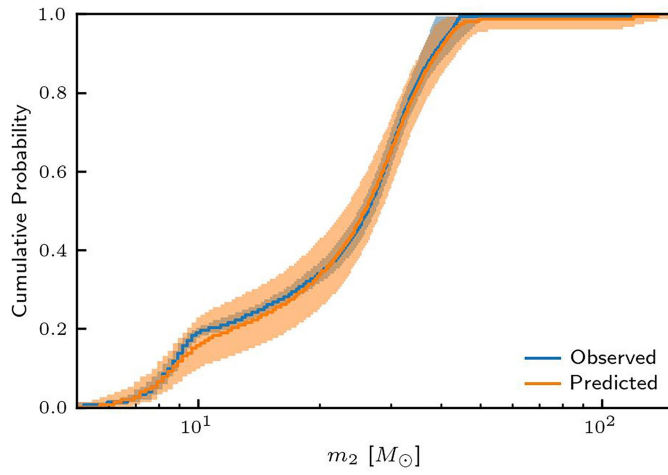
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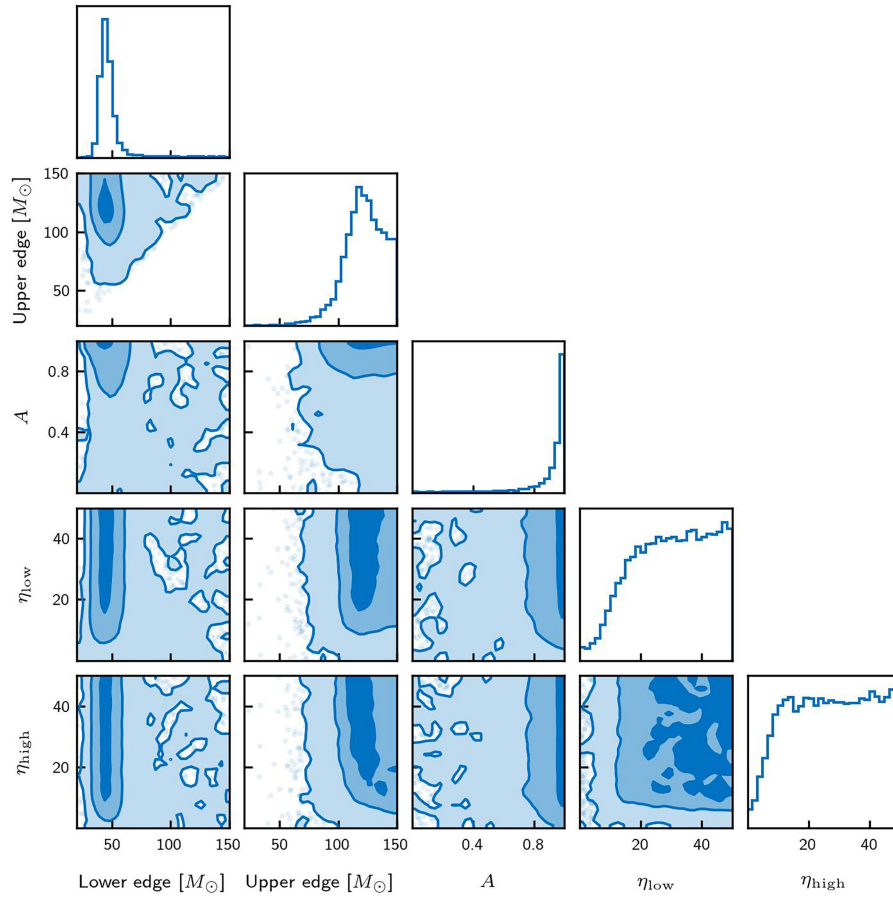
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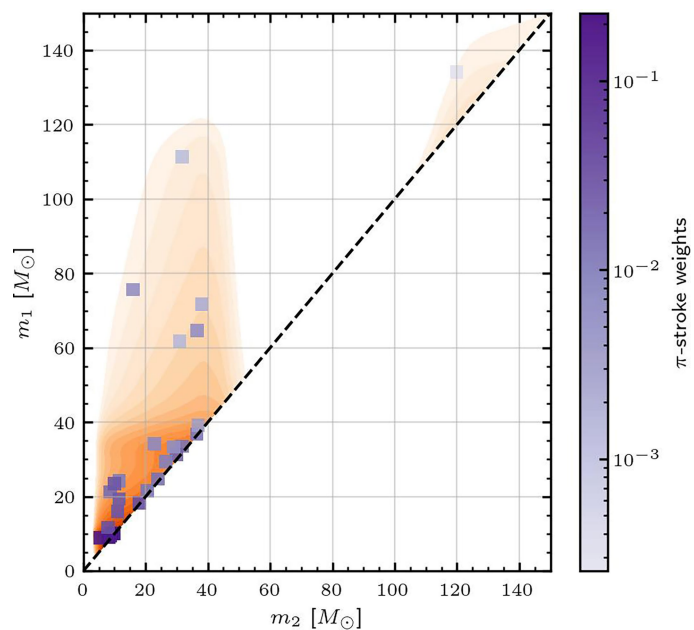
Extended Data Fig. 1 | Corner plots of the lower and upper boundaries and the width of the secondary mass gap. The results from GWTC-4 including the exceptionally massive event GW231123 are shown in blue. This event is necessary to obtain a useful constraint on the upper edge of the mass gap. Constraints on the lower edge obtained with GWTC-4 excluding GW231123 (orange) and with GWTC-3 (green) are made by inferring the maximum truncation in the m_2 distribution. The constraints on the lower edge excluding GW231123 are consistent with those obtained with GW231123.



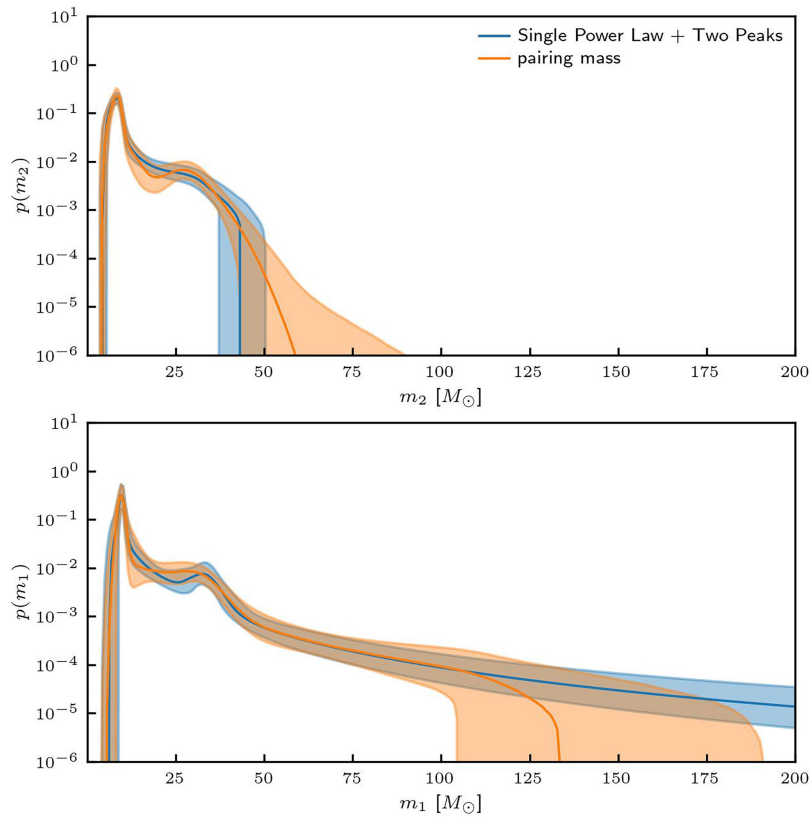
Extended Data Fig. 2 | A posterior predictive check. The plot shows the cumulative function of the observed events' secondary-mass distribution (blue) versus the predicted secondary-mass distribution for the single power law + two peaks model with a gap in m_2 (orange). The solid lines indicate the median and the shaded bands indicate the 90% credibility range.



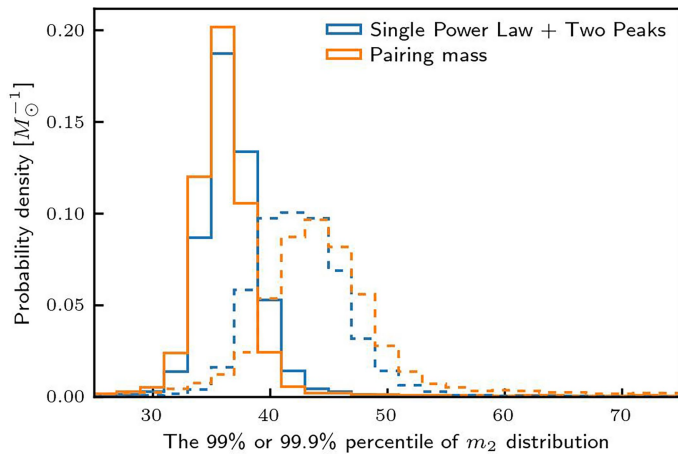
Extended Data Fig. 3 | Corner plots of the notch-filter parameters. The plots show the posterior distribution of the parameters that describe the notch filter in equation (4). The contours in the 2D panels indicate the 1σ , 2σ and 3σ credible intervals.



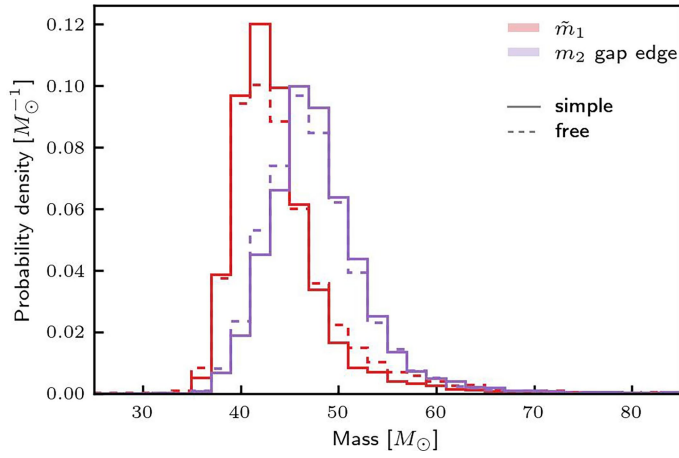
Extended Data Fig. 4 | Predicted population distribution by the maximum population likelihood method. The distribution of the maximum population likelihood is expressed as a weighted sum of delta functions, using the π -stroke (π) formalism. Scatter points show where the delta functions are in terms of (m_1, m_2) . The colour bar represents the weights of delta functions. We also show the mean prediction of the joint distribution $\pi(m_1, m_2)$ by our standard parametric model analysis in the background for reference. Both analyses show consistent support for the m_2 gap.



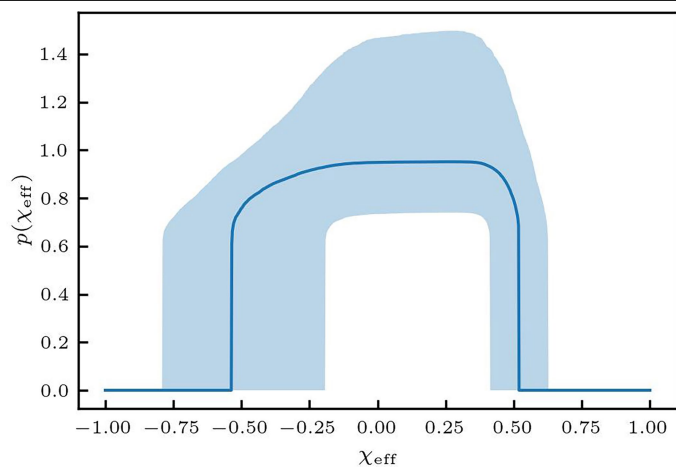
Extended Data Fig. 5 | Reconstructed distributions of primary masses m_2 (top) and secondary masses m_1 (bottom). The blue fit is obtained with the single power law + two peaks model and the orange fit is obtained with the pairing-mass model from ref. 79. The shaded bands indicate the 90% credibility ranges.



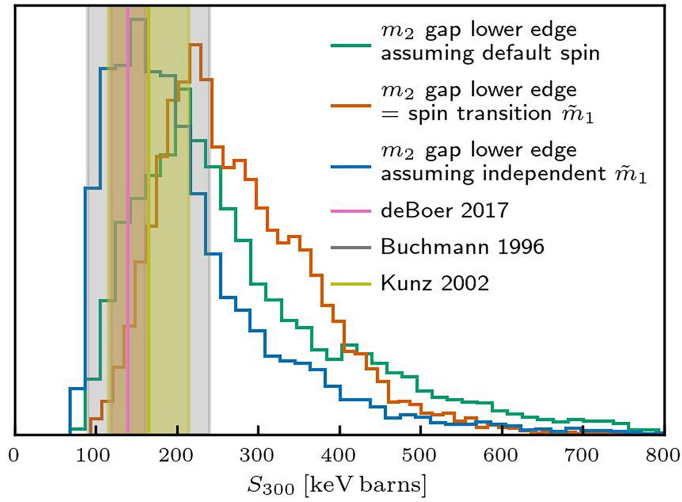
Extended Data Fig. 6 | Comparing quantiles for different models. 99% of mergers are below $m_2^{99\%}$ (solid lines) and 99.9% are below $m_2^{99.9\%}$ (dashed-dotted lines). Blue shows the results for the single power law + two peak model and orange shows the results for the more flexible pairing-mass model⁷⁹.



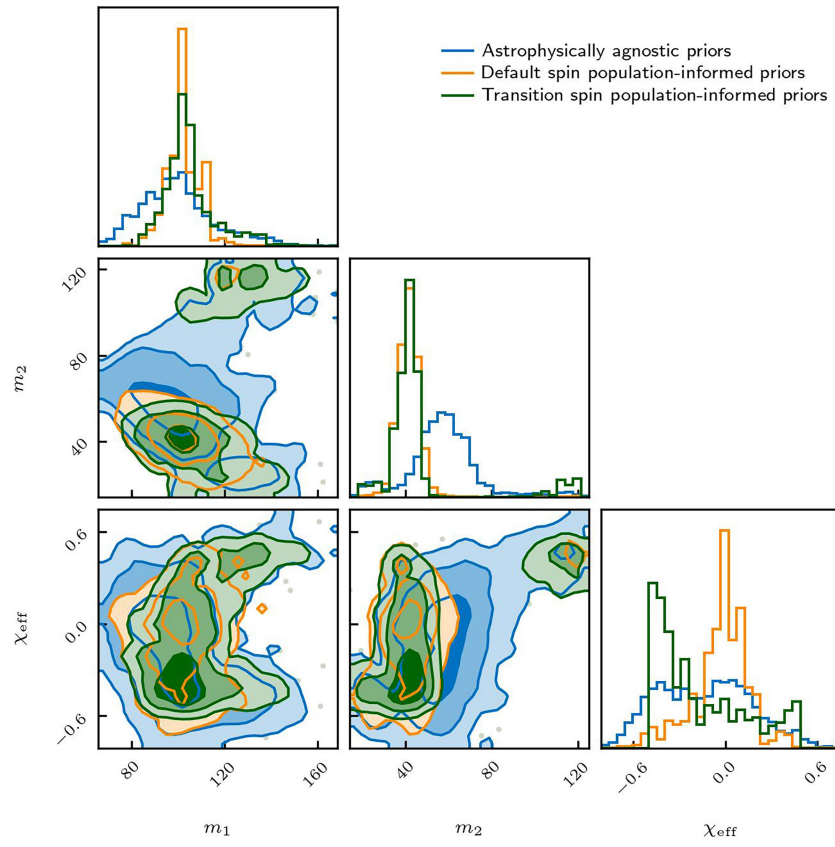
Extended Data Fig. 7 | Comparing the lower edge of the mass gap and the spin-transition mass results using different spin models. We show the results as solid lines assuming the ‘simple’ spin-transition model from equation (7), whereas the results shown by dashed-dotted lines assume the more complicated spin-transition model from equation (8). In purple is the posterior for the lower edge of the m_2 mass gap. Red shows the posterior for the spin-transition mass $\tilde{m}_1$. The results are consistent between different spin-transition models.



Extended Data Fig. 8 | The effective spin distribution for high-mass binary black holes with $m_1 > \bar{m}_1$ using the flexible spin model in equation (9). The shaded area represents the 90% credible interval and the solid line is the median.



Extended Data Fig. 9 | Constraints on the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate with various population model assumptions. Green is the posterior for S_{300} using the lower edge of the mass gap inferred assuming the default spin model in ref. 40. Blue uses the measurement of the lower edge of the mass gap assuming the spin transition model in ref. 21, whereas the spin transition mass $\tilde{m}_1$ is independent and non-identical to the m_2 gap. Orange is the posterior using the measurement imposing the lower edge of the m_2 gap to be equal to the spin transition mass $\tilde{m}_1$. Results in pink, grey and yellow-green show the theoretical nuclear physics predictions from refs. 45–47.



Extended Data Fig. 10 | The inferred population-informed estimates of the primary mass m_1 , secondary mass m_2 and effective spin χ_{eff} for GW190521. The contours represent 1σ , 2σ and 3σ credible regions, respectively.

Extended Data Table 1 | Model selection results

Model	$\ln \mathcal{B}$	Relative maximum $\ln \mathcal{L}$
Broken Power Law + Two Peaks without m_2 gap	1	0
Single Power Law + Two Peaks with m_2 gap	4.4	8.84
Broken Power Law + Two Peaks with m_2 gap	3.7	9.23
Single Power Law + Two Peaks with m_2 notch filter	1.8	6.98
Single Power Law + Two Peaks with m_2 notch filter (A=1)	4.4	-

The natural logarithm Bayes factor and maximum likelihood comparison of several variations of the basic mass model with the default model of ref. 40 as the baseline. The Bayes factor of single power law+two peaks with the m_2 notch-filter model assuming A=1 (that is, empty gap case) is approximated by the Savage–Dickey density ratio.